A Short Note on Attention-Based Sequence Models

Abstract

This sample note describes a sequence model built entirely from attention layers. Instead of reading tokens one at a time, the model lets every token look at every other token in parallel. The main contribution is showing that this design trains faster than recurrent models and reaches higher translation quality on the benchmark used here. This file is a placeholder so the pipeline can be tested; replace it with a real paper.

1. Introduction

Recurrent networks process a sequence step by step, which makes training hard to parallelize and makes long-range dependencies difficult to learn. The model proposed in this note removes recurrence completely. Each layer relies on self-attention, a mechanism where every position in the input computes a weighted combination of all other positions.

2. Model Architecture

The model has an encoder and a decoder. The encoder stacks six identical layers, each containing a multi-head self-attention block followed by a feed-forward network. The decoder has the same structure plus a block that attends to the encoder output. Because attention has no built-in notion of order, positional encodings are added to the token embeddings.

3. Multi-Head Attention

Attention uses queries, keys and values. The similarity between a query and each key is scaled and passed through a softmax to produce weights, which are then used to average the values. Multi-head attention runs eight of these operations in parallel with different learned projections, letting the model focus on different kinds of relationships at once.

4. Training

The model was trained on a translation dataset using the Adam optimizer with a warmup learning-rate schedule. Dropout of 0.1 was applied to every sub-layer. Training took about twelve hours on eight GPUs, a fraction of the time needed by the recurrent baselines.

5. Results

On the English-to-German test set the model scored 28.4 BLEU, improving on the best recurrent system by more than two points. On English-to-French it scored 41.0 BLEU. The authors attribute the gains to the ability to model long-range dependencies directly.

6. Limitations

Self-attention cost grows quadratically with sequence length, so very long inputs are expensive. The note only evaluates translation, so conclusions about other tasks are untested.

7. Conclusion

Attention alone is enough to build a strong sequence model. It is faster to train, easier to parallelize, and more accurate than the recurrent baselines evaluated here.